

Md Mahmuddun Nabi Murad

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Ph.D. Candidate in Electrical Engineering at the University of South Florida, specializing in Deep Learning, Time-Series Forecasting, and Anomaly Detection.

Professional Summary

Machine Learning Researcher with a Ph.D. focus on designing advanced neural architectures (Transformers, MLP-Mixers, GNNs). Proven track record of developing SOTA models for long-term forecasting and sequential data, evidenced by a first-author publication at AAAI and winning a global data competition. Expert in Python and the PyTorch ecosystem, with a strong foundation in signal processing and cross-domain model adaptation.

Technical Skills

Languages: Python, C/C++

ML/DL: PyTorch, TensorFlow (familiar), scikit-learn, NumPy, Pandas

Tools & Platforms: Git, Linux, LaTeX, Optuna, MATLAB

Experience

Graduate Research Assistant

Aug 2023 – Present

SISLab, University of South Florida

- **Neural Architecture Design:** Developed **WPMixer**, a novel MLP-mixer-based model for multi-resolution time-series forecasting, outperforming Transformer variants in accuracy and efficiency (**AAAI 2025**).
- **Anomaly Detection:** Developed a novel causal MLP-mixer model with cluster-based embeddings to detect sequential anomalies in high-dimensional multivariate time series with a sequential anomaly detection framework.
- **Sensor Localization:** Designed a multi-resolution mixer network for sensor localization from RF power measurements, significantly reducing error rates compared to classical approaches.
- **Computer Vision:** Implemented a real-time intruder detection system with work-hour estimation of the crew members for maritime environments, integrating object detection with behavioral monitoring.

Protection Engineer (Data & Systems Analysis)

Dec 2019 – Jul 2023

Power Grid Bangladesh PLC

- Performed complex **fault analysis** across the national grid, utilizing statistical methods to identify root causes of system failures in high-frequency data streams.
- Configured Substation Automation Systems (SAS) to manage real-time monitoring and protection coordination for critical infrastructure.
- Translated physical performance metrics into data-driven maintenance schedules through rigorous equipment testing.

Key Machine Learning Projects

Global Hackathon Winner: Vessel Route Prediction (M3 Hackathon):

(Organized by the U.S. Department of Homeland Security.)

- Developed a hybrid **Transformer-MLP-Mixer** architecture to predict vessel trajectories using AIS data; secured **1st Place** globally. Additionally, implemented a custom endpoint detection algorithm.

Time-Series Forecasting:

- Developed a novel long-term multivariate time series forecasting model with wavelet-based decomposition and reconstruction. The model outperforms SOTA models, including Transformer-based variants.

Time-Series Anomaly Detection:

- Developed a sequential anomaly detection framework, with a novel causal mixer and cluster-based embeddings in the reconstruction model.

Soil Moisture Estimation and Sensor Localization via RF Signals:

- Estimated soil moisture and localized sensors from the *cumulative received signal strength*. A stationary interrogator transmits to distributed nodes that reflect signals back; the interrogator observes the aggregate power and infers moisture and positions. Nodes are deployed in soil and may drift with natural events (e.g., landslides), enabling joint tracking and analysis.

Video-Based Sequential Route Prediction:

- Engineered a **Computer Vision pipeline** for vessel trajectory prediction, transitioning from AIS-based modeling to visual sequential data analysis.
- **Curated a novel maritime video dataset** specifically designed to address data sparsity and benchmark model performance in complex, real-world visual environments.

Research Publications

[1] **M. M. N. Murad**, M. Aktukmak, and Y. Yilmaz, “Wpmixer: Efficient multi-resolution mixing for long-term time series forecasting,” in *Proceedings of the AAAI conference on artificial intelligence*, 2025, arXiv preprint arXiv:2412.17176.

[2] **M. M. N. Murad** and Y. Yilmaz, “Cluster-aware causal mixer for online anomaly detection in multivariate time series,” 2025, under review. [Online]. Available: <https://arxiv.org/abs/2506.00188>

[3] **M. M. N. Murad**, D. K. Yirenya-Tawiah, T. Weller, and Y. Yilmaz, “Multi-resolution mixer network for localization of multiple sensors from cumulative power measurements,” in *IEEE Wireless Communications and Networking Conference (WCNC)*, 2025, accepted (Paper ID: 1571076702).

[4] **M. M. N. Murad**, B. S. Turgut, A. Ahmed, G. Camliyurt, and Y. Yilmaz, “Camera-based intruder detection and monitoring of ship crew work hours,” in *Proceedings of the Winter Conference on Applications of Computer Vision*, 2025, accepted (Paper ID: 13).

Education

Ph.D. in Electrical Engineering (ML Focus) <i>University of South Florida (USF), Tampa, FL</i> CGPA: 4.0/4.0	Aug 2023 – Present
B.Sc. in Electrical and Electronic Engineering <i>Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh</i> CGPA: 3.67/4.0	Feb 2015 – Apr 2019

Awards and Prizes

Managing Maritime Movements (M3) Hackathon <i>M3 is a global hackathon on small vessel route prediction. I developed a novel Transformer–MLP-Mixer hybrid model for route prediction and proposed a route’s endpoint detection method.</i>	Winner (First Prize), 2024
BUET Dean’s list award <i>Achieved the prestigious dean’s list award for outstanding academic performance in one semester during undergraduate studies.</i>	

Selected Coursework

EEL 6935 Advance Data Analytics	ESI 6684 Decision Making with Deep RL
EEE 6777 Data Analytics	EEL 6935 Linear and Matrix Algebra
EEL 6789 Deep Learning	EEL 6935 Advanced GenAI

References

Available upon request.